

and elfnchips.

While hiring a VLSI engineer, these companies look for strong fundamentals in basic electronics and semiconductor concepts like system design, timing and semiconductor physics. They also look for good knowledge of languages like VHDL, Verilog, Specman and System Verilog, as well as good programming skills and scripting abilities. In lateral hires, they look for strong domain-understanding or sector-specific knowledge, as the industry is moving towards vertical offerings in chip design.

Surinder Bhagat, country HR manager, Freescale Semiconductor India, says, "We prefer an educational background of BE/B.Tech, ME/M.Tech or PhD with specialisation in electronics, telecommunications, electrical or VLSI domain. Typically, these professionals should have hands-on experience in systems design, digital ASIC design, physical design, mixed-signal IC design, VHDL or very high-speed integrated circuit, VLSI design, circuit design and simulations, microcontrollers, digital PCB design and routing."

The challenges

Going by the projections of high demand and shortage of manpower, it is the right time to enter this industry and get benefits in the future. According to Dr Subbarangaiah, director, VEDA IIT, "Even though around 250,000 engineers are graduating in electronics and other related streams every year, less than 3 per cent of them are equipped with the knowledge and skills needed for the VLSI industry."

To add to this, the demand is constantly increasing. There is an acute shortage of skilled engineers, particularly for physical design and analogue design. With the technology advancing at a fast pace, it makes sense to undergo technical training in working with nanotechnology to become industry-ready. The training institute should have access to advanced technology libraries, designs, industry-standard sign-off EDA tools

and design flows. There are not many institutes that can train the students in such niche areas in spite of their special manpower development programme.

Says Bhagat, "We are constantly looking to hire both fresh and experienced talent to meet demand. If academia can help the industry by introducing courses designed to help freshers become productive right from day one in the company, it can help the industry as the employer saves six to nine months, which otherwise are spent on training. It is very important that the industry and academia come together on a platform where they can exchange dialogue and design the right curriculum for the students."

Kathuria suggests a way forward: "A person who wishes to design ASICs will require extensive training in the field of VLSI design. But we cannot possibly expect that a large number of people would wish to undergo such training. Also, the process of training these people will itself entail large investments in terms of time and money. This means there has to be a system that can abstract out all the details of VLSI, and allows the user to think in simple system-level terms."

How much remuneration to expect?

The industry offers good compensation to experienced candidates. Generally, salaries are based on the individual skillsets and qualification. According to Aaghoramoorthy, this is a market known for dynamic shifts in technology and hence engineers with experience in working with newer design and verification methodologies are in high demand.

In the long run, engineers who have built strong domain skills and possess strong fundamental knowledge of circuit, logic design or verification are valued, since they can quickly learn the electronics tool related aspects.

"An engineer starting his career with a bachelor's degree can expect an annual package of Rs 400,000 to 700,000. Many companies give higher salaries to engineers with M.Tech or PhD based on their continual performance, knowledge and competence. One can expect a steep growth in remuneration as well as roles and responsibilities. For example, a fresher can easily move up to become a team leader or project manager in a span of seven to ten years. Typically, fresh college graduates join as a design engineer and continue to grow their skill level over next six to twelve months in design methodology, along with working on other design projects," says Bhagat.

Agarwal classifies the pay packages according to recruiters from the embedded industry: "Companies which are large in terms of revenue or client base and have already set a global landmark—for example, Intel, Tata Elxsi, Synopsys, IBM, Xilinx, Windriver and Texas Instruments—offer anywhere between Rs 400,000 and 500,000 per annum to fresh engineering graduates. Other companies having a decent revenue inflow—for example, Coreel Technologies, Moschip, Radel, PACE, elfnchips and Arada Systems—offer anywhere between Rs 250,000 and 350,000 per annum. Another category of companies comprises small set-ups or start-ups that are catching up well with the market—for example, Riversilica, Fossilshale, Saankhya, Whizchip and Accord. These offer anywhere between Rs 150,000 and 250,000 per annum, depending on the calibre of the candidate."

Hence there is no doubt that with all the innovation and rapid development, this field offers you ample scope to grow. So if you already enjoy the brain teasers in designing and testing the chips, get set for the strong growth promised by the VLSI industry. ●

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